

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech. Branch : Computer Engineering Semester :IV

Subject Code & Name: BTCOC403 Operating System

Max Marks: 60 Date: 20/01/2024 Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q. 1 Solve Any Two of the following. 12

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| A) Explain the Importance of Real-Time Embedded systems. | Understand | 6 |
| B) Define Operating System. List the objectives of an operating system. | Apply | 6 |
| C) What is a Virtual Memory? Discuss the benefits of virtual memory technique. | Apply | 6 |

Q.2 Solve Any Two of the following. 12

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|---|------------|---|
| A) What is a process? Explain about various fields of Process Control Block. | Understand | 6 |
| B) Using the given information about the process, calculate average waiting time and average turnaround time of each process under the following scheduling algorithm: 1)FCFS, 2)SJF 3) SJF (Preemptive mode) | Apply | 6 |

Process Id	Arrival Time	Burst Time
P1	0	8
P2	1	5
P3	3	4
P4	4	3
P5	1	3

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|--|------------|---|
| C) Explain the steps involved in process creation and process termination. | Understand | 6 |
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Q. 3 Solve Any Two of the following. 12

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| A) What is Dining Philosophers problem? Discuss the solution to Dining philosopher's problem using monitors. | Understand | 6 |
| B) What are the semaphores? How do they implement mutual exclusion? | Understand | 6 |
| C) What is a scheduler? List and describe different types of schedulers. | Understand | 6 |

Q.4	Solve Any Two of the following.		12
A)	How does deadlock avoidance differ from deadlock prevention? Write about deadlock avoidance algorithm in detail.	Understand	6
B)	Given memory partition of 100 KB, 500 KB, 200 KB and 600 KB (in order). Show with neat sketch how would each of the first-fit, best-fit and worst fit algorithms place processes of 412 KB, 317 KB, 112 KB and 326 KB (in order).	Apply	6
C)	Write a short note on the following:	Understand	6
	i) Stable storage implementation ii) Free space management		
Q. 5	Solve Any Two of the following.		12
A)	Consider the reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 for a memory with three frames. Trace FIFO, optimal, and LRU page replacement algorithms. https://www.batuonline.com	Apply	6
B)	How to provide protection to a file system? Explain.	Understand	6
C)	Discuss the different file allocation methods with suitable example	Analysis	6
*** End ***			